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Roll No

ME-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Dynamics of Machines

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data or dimensions, if necessary.

1. a) Derive a relation for the turning moment at the crank shaft in the terms of piston effort and angle turned by the crank.
b) A flywheel is used to give up 18kJ of energy in reducing its speed from 100rpm to 98rpm. Determine its kinetic energy at 140rpm.
2. a) What is meant by effort and Power of a governor? Find the expression for the same in a Hartnell governor.
b) A porter governor has equal arms each 200mm in length and pivoted on the axis of rotation. The mass of the each ball is 4kg and mass of the sleeve is 20kg. The radius of the rotation of the ball is 100mm when the governor begins to lift. If the fractional increase of speed is 1%, then determine the governor effort and power of the governor.
3. a) Define and explain the balancing of rotation masses. What will be the harm if the rotating parts of a high speed engine are not properly balanced?
b) Explain the method of direct and reverse cranks to determine the unbalance forces in radial engines.
4. a) Describe with neat sketch the working of the single plate friction clutch.

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- b) Prove that the angle of friction is equal to the angle of the inclined plane, when a solid body of the weight W placed on the inclined plane is about to slide down.
- 5. a) What do you understand by 'Self-locking brake' and 'Self-energized brake'?
- b) What is the principle on which a torsion dynamometer works? Explain with details the calculation involved in finding the power transmitted.
- 6. A shaft is rotating at a uniform angular speed. Four masses m_1 , m_2 , m_3 and m_4 of magnitude 300kg, 450kg, 360kg and 390 kg respectively are attached rigidly to the shaft. The masses are rotating in the same plane. The corresponding radii of rotation are 200mm, 150mm, 250mm and 300mm respectively. The angle made by these masses with horizontal is 0° , 45° , 120° and 255° respectively. Determine the magnitude of the balancing mass and the position of the balancing mass if the radius of rotation is 200mm.
- 7. ABCD is a 4-bar chain with AB as the driving link and AD as the fixed link. The length of link AB = 7.5cm, BC = 17.5cm, CD = 15cm and AD = 22.5cm. The diameters of the pin at A, B, C, and D are all 2cm. What torque must be applied to the driving shaft in order to overcome a resisting torque of 3kgf applied to the shaft to which CD is keyed, when the angle BAD = 90° , take $\mu = 0.125$.
- 8. Write short notes on following (Any four);
 - a) Hunting of governor
 - b) Lanchester technique
 - c) Boundary and fluid film lubrication
 - d) Antifriction bearings
 - e) Jump and cross over shock
 - f) Friction circle

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